Tech Bite #11: Working Amateur Satellites

Yes -- a Technician can talk through satellites. How amateur satellites work, and what you need to know to use one.

Group T8B · 12 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

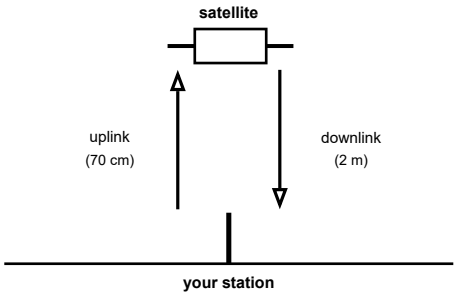
The Big Picture

Here is one of the genuinely thrilling things about this license: as a **Technician, you can make contacts through satellites**. This Bite covers how amateur satellites work and what you need to know to use one. No special equipment beyond a dual-band radio and a small antenna is required to get started.

Uplink and Downlink

A satellite contact has two paths: the **uplink** (your signal up to the satellite) and the **downlink** (the satellite's signal back down). When a satellite is in **"U/V mode,"** the **uplink is on 70 centimeters** and the **downlink is on 2 meters**. (T8B08)

Satellites use whatever modes suit them – **SSB, FM, CW, and data**. (T8B04) Every amateur satellite sends a **beacon**: a transmission carrying its **health-and-status telemetry** – and **anyone** may receive it. (T8B01, T8B05, T8B11)



A satellite contact: your uplink goes up, the downlink comes back – on different bands.

Doppler Shift and Spin Fading

Because a satellite moves fast relative to you, its signal frequency seems to drift – this is **Doppler shift**, an apparent change in frequency caused by the **relative motion** between the satellite and your station. (T8B07) And because the satellite tumbles, its signal can rise and fall – **spin fading**, caused by the **rotation of the satellite and its antennas**. (T8B09)

Finding and Working a Satellite

Most amateur satellites are in **LEO** – **Low Earth Orbit** – with a period of about **100 minutes**. (T8B10) A **satellite tracking program** tells you when and where to look: it provides maps, pass times, and the Doppler-shifted frequency, using the satellite's **Keplerian elements** as input. (T8B03, T8B06) Two power rules: do **not** use **excessive uplink power** – it **blocks other users** (T8B02) – and on a **linear-transponder** satellite, set your power so your **downlink signal is about as strong as the satellite's beacon**. (T8B12)

Mental Cheat Sheet

- A contact has an **uplink** (up) and a **downlink** (down).
- **U/V mode**: uplink 70 cm, downlink 2 m.
- A **beacon** sends the satellite's health/status; **anyone** may receive it.
- **Doppler shift** = apparent frequency change from relative motion.
- **Spin fading** = signal rising and falling as the satellite rotates.
- **LEO** = Low Earth Orbit, period about 100 minutes.
- Tracking programs use **Keplerian elements** as input.
- Do not use excessive uplink power – it **blocks others**.

Common Traps

- In **U/V mode**, U = uplink on 70 cm, V = downlink on 2 m.
- **Doppler shift** is from motion – not a change in the orbit itself.
- **Spin fading** is the satellite **rotating** – not Doppler, not the sun.
- **LEO** is Low **Earth** Orbit – not "Low Energy" or "Low Elevation."
- **Anyone** may receive satellite telemetry – no key or license needed to listen.
- On a linear transponder, match your downlink to the **beacon** strength.

Takeaway: Working a satellite is a real Technician privilege. A contact has an **uplink** and a **downlink** – often 70 cm up and 2 m down (U/V mode). The satellite **moves**, so expect **Doppler shift** and **spin fading**; a **tracking program** tells you when it passes overhead. Keep your **uplink power modest** so you do not block other users – and enjoy one of the most fun things on the bands.

Bite 11 Question Bank (T8B)

T8B01. What telemetry information is typically transmitted by satellite beacons?
A. The signal strength of received signals
B. Time of day accurate to plus or minus 1/10 second
C. Health and status of the satellite
D. All these choices are correct

T8B02. What is the impact of using excessive effective radiated power on a satellite uplink?
A. Possibility of commanding the satellite to an improper mode
B. Blocking access by other users
C. Overloading the satellite batteries
D. Possibility of rebooting the satellite control computer

T8B03. Which of the following are provided by satellite tracking programs?
A. Maps showing the real-time position of the satellite track over Earth
B. The time, azimuth, and elevation of the start, maximum altitude, and end of a pass
C. The apparent frequency of the satellite transmission, including effects of Doppler shift
D. All these choices are correct

T8B04. What mode of transmission is commonly used by amateur radio satellites?
A. SSB
B. FM
C. CW/data
D. All these choices are correct

T8B05. What is a satellite beacon?
A. The primary transmit antenna on the satellite
B. An indicator light that shows where to point your antenna
C. A reflective surface on the satellite
D. A transmission from a satellite that contains status information

T8B06. Which of the following are inputs to a satellite tracking program?
A. The satellite transmitted power
B. The Keplerian elements
C. The last observed time of zero Doppler shift
D. All these choices are correct

T8B07. What is Doppler shift in reference to satellite communications?
A. A change in the satellite orbit
B. A mode where the satellite receives signals on one band and transmits on another
C. An observed change in signal frequency caused by relative motion between the satellite and Earth station
D. A special digital communications mode for some satellites

T8B08. What does it mean if a satellite is operating in U/V mode?
A. The satellite uplink is in the 15-meter band and the downlink is in the 10-meter band
B. The satellite uplink is in the 70-centimeter band and the downlink is in the 2-meter band
C. The satellite operates using ultraviolet frequencies
D. The satellite frequencies are usually variable

T8B09. What causes spin fading of satellite signals?
A. Circular polarized noise interference radiated from the sun
B. Rotation of the satellite and its antennas
C. Doppler shift of the received signal
D. Interfering signals within the satellite uplink band

T8B10. What does the term LEO mean in reference to communication satellites?
A. Low Energy Orbit, which conserves battery power
B. Low Elevation Orbit, which appears close to the horizon from the earth station
C. Low Equilibrium Orbit, which has a slightly unstable period
D. Low Earth Orbit, which has a period of around 100 minutes

T8B11. Who is permitted to receive telemetry from an amateur radio satellite?
A. Anyone
B. Only the satellite control operator
C. Only the control operator or a licensed radio amateur who has received the encryption key from the control operator
D. Only a licensed radio amateur who has received the encryption key from AMSAT

T8B12. Which of the following is a way to determine whether your satellite uplink power into a linear transponder satellite is neither too low nor too high?
A. Check your signal strength report in the telemetry data
B. Listen for distortion on your downlink signal
C. Your signal strength on the downlink should be about the same as the beacon
D. Compare your signal to others on the downlink using an internet SDR receiver

Answer Key

T8B01: C	T8B02: B	T8B03: D	T8B04: D
T8B05: D	T8B06: B	T8B07: C	T8B08: B
T8B09: B	T8B10: D	T8B11: A	T8B12: C